

25-12-05_update

Default Clustering behavior

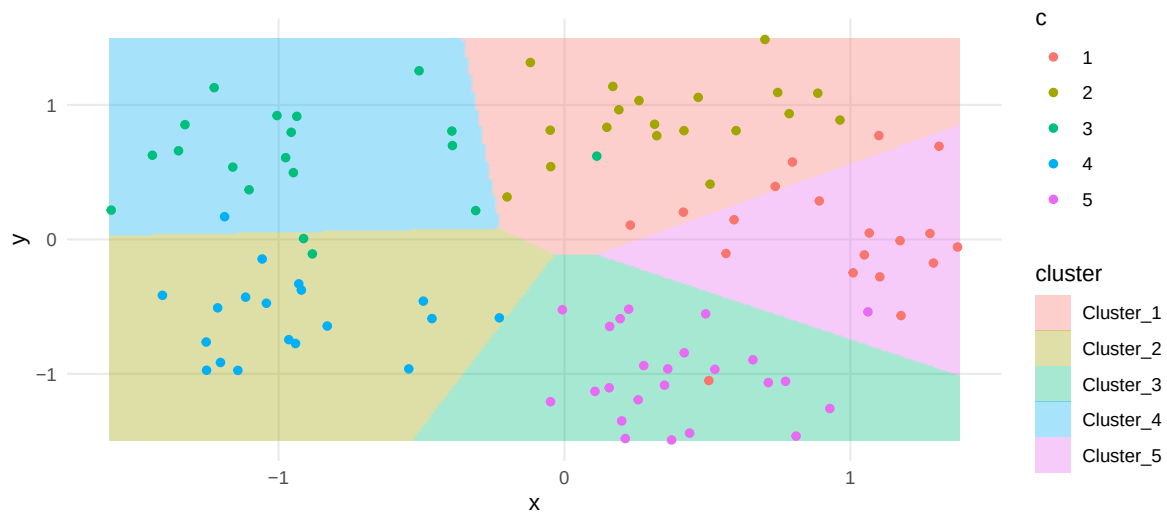
The available k means engines available with tidyclust require picking some number of clusters so the previous default was a rather arbitrary $k = 3$ k means model if the user didn't supply a method.

To get a stronger default the new behavior if no method is supplied is 5 fold cross validation across $k = 2$ to 10, where the k with the highest average silhouette value is selected. The user is warned this was selected and that value is used to plot.

This does require tidymodels to use.

```
samp <- rnorm_mixture(100, mean = roots_of_unity(5), var = .1)
df <- data.frame(x = samp[,1], y = samp[,2], c = attr(samp, "component"))
ggplot(df, aes(x, y))+
  geom_cluster()+
  geom_point(aes(col = c))
```

! a k means model with 5 clusters was used



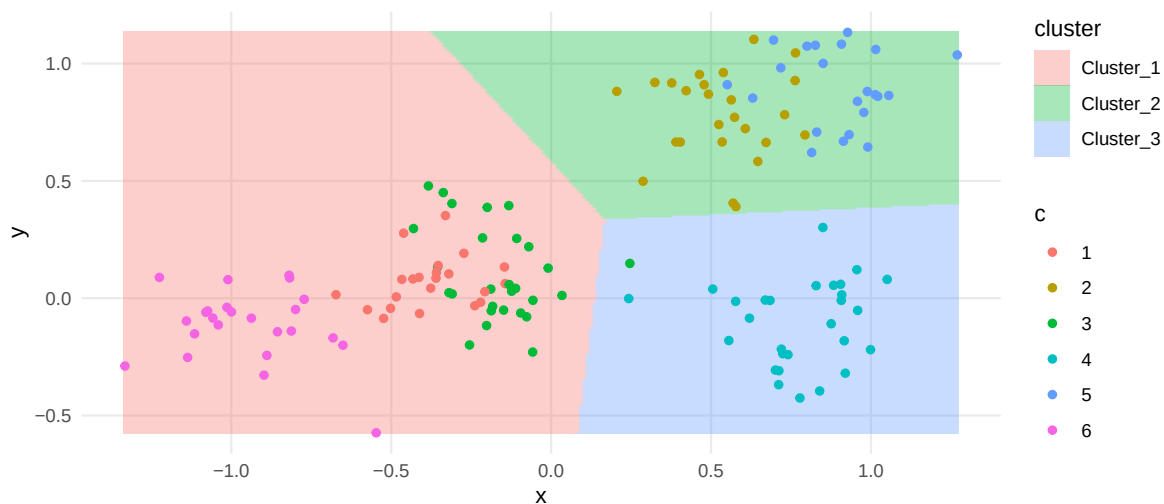
The model is pretty effective with Roots of Unity

```

set.seed(123)
means <- cbind(x = runif(6,-1), y = runif(6,-1))
samp <- rnorm_mixture(149, mean = means, var = .025)
bidata <- tibble(x = samp[,1], y = samp[,2], c = attr(samp, "component"))
ggplot(bidata, aes(x, y))+
  geom_cluster()+
  geom_point(aes(col = c))

```

! a k means model with 3 clusters was used



More complex data can be difficult though.

I have my method for selecting k below. Not the most experience with building tidymodels so their could be inefficiencies.

```

get_best_kmeans <- function(data){

  data_cv <- vfold_cv(data, v = 5)

  data_rec <- step_normalize(recipe(~ x + y, data = data), all_predictors())

  kmeans_spec_tune <- set_engine(k_means(num_clusters = tune()), "stats", iter.max = 25, nst

  kmeans_wflow <- add_model(add_recipe(workflow(), data_rec), kmeans_spec_tune)

  clust_metrics <- cluster_metric_set(silhouette_avg)

```

```

data_tune <- tune_cluster(
  kmeans_wflow,
  resamples = data_cv,
  grid = tibble(num_clusters = 2:10),
  metrics = clust_metrics
)

metrics_tbl <- collect_metrics(data_tune)

best_k <- metrics_tbl$num_clusters[which.max(metrics_tbl$mean)]
cli_alert_warning(paste0("a k means model with ", best_k, " clusters was used"))
set_engine(k_means(num_clusters = best_k), "stats", iter.max = 40, nstart = 10)
}

```

```

datasets::penguins |> remove_missing() |> mutate(x = bill_len, y = bill_dep) |>
ggclassify::get_best_kmeans()

```

! a k means model with 3 clusters was used

K Means Cluster Specification (partition)

Main Arguments:

num_clusters = best_k

Engine-Specific Arguments:

iter.max = 40

nstart = 10

Computational engine: stats

Documentation work

Man pages

I have condensed the documentation from `geom_classify`, `geom_classify_boundary`, and `geom_point_classify` into one page. as well as `geom_cluster`, `geom_cluster_boundary`, and `geom_point_cluster` into one page. This should help reduce the amount of copy paste documentation. I've further outlined the computed stats available to each geom.